

No subspace counterexample to the $\mathfrak{F}_{cmc}$ -SACP sum question

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Abstract

Das asks in Remark 3.12 of [1] whether two subspaces Y, Z of a Banach space X can each have τ - $\mathfrak{F}_{cmc}$ -SACP while $Y + Z$ is closed and $Y + Z$ fails τ - $\mathfrak{F}_{cmc}$ -SACP. The answer is negative for both topologies allowed in the paper, namely norm topology and weak topology. The reason is that the full class $\mathfrak{F}_{cmc}$ contains the plateau gauge $t \mapsto \max\{t - 1, 0\}$. Applied to the one-point set $\{0\}$, this gauge turns every sequence in the unit ball of a subspace into a minimizing sequence. Consequently norm- $\mathfrak{F}_{cmc}$ -SACP forces finite dimensionality, and weak- $\mathfrak{F}_{cmc}$ -SACP forces reflexivity. The finite-dimensional and reflexive transfer theorems already proved in [1] then rule out the requested subspace counterexample.

1 Source question and definitions

The source paper fixes a real Banach space X , declares that subspaces are closed linear subspaces, and lets τ denote either the norm topology or the weak topology on X . For a τ -closed subset $V \subset X$, a nonempty finite set $F = \{x_1, \dots, x_N\} \in \mathcal{F}(X)$, and a function $f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+$, it defines

$$r_f(x, F) = f(\|x - x_1\|, \dots, \|x - x_N\|), \quad \text{rad}_V^f(F) = \inf_{v \in V} r_f(v, F).$$

The triplet $(X, V, \mathcal{F})$ has τ - $\mathfrak{F}$ -SACP if every sequence $(v_n) \subset V$ satisfying

$$r_f(v_n, F) \longrightarrow \text{rad}_V^f(F)$$

has a τ -convergent subsequence, for every allowed finite set F and every allowed f .

The function class used in the source paper is

$$\mathfrak{F}_{cmc} = \bigcup_{N \in \mathbb{N}} \{f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+ : f \text{ is convex, monotone, and coercive}\}.$$

Here monotone means coordinatewise monotone, and coercive means $f(\varphi) \rightarrow \infty$ as $\|\varphi\|_\infty \rightarrow \infty$.

Remark 3.12 asks:

“We do not know whether there exist two subspaces Y and Z of X such that $(X, Y, \mathcal{F}(X))$ and $(X, Z, \mathcal{F}(X))$ has τ - $\mathfrak{F}_{cmc}$ -SACP, $Y + Z$ is closed but $(X, Y + Z, \mathcal{F}(X))$ does not have τ - $\mathfrak{F}_{cmc}$ -SACP.”

characterizations of these properties. Throughout this paper, we restrict the class $\mathcal{F}$ to the following family of functions

$$\mathcal{F}_{cmc} := \bigcup_{N \in \mathbb{N}} \{f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+ \mid f \text{ is convex, monotone, and coercive}\},$$

where $f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+$ is said to be

- (a) *monotone* if for $\varphi_1, \varphi_2 \in \mathbb{R}_+^N$, $\varphi_1 \leq \varphi_2$ implies $f(\varphi_1) \leq f(\varphi_2)$, where $\varphi_1 \leq \varphi_2$ if and only if $\varphi_1(i) \leq \varphi_2(i)$ for all $i = 1, \dots, N$,
- (b) *coercive* if $f(\varphi) \rightarrow \infty$ as $\|\varphi\|_\infty \rightarrow \infty$.

Figure 1: The source definition of $\mathfrak{F}_{cmc}$.

τ -convergent subsequence in ℓ_1 . Thus $(X, U + V, \mathcal{F}(X))$ does not have τ - $\mathcal{F}_{cmc}$ -SACP.

Remark 3.12. We do not know whether there exist two subspaces Y and Z of X such that $(X, Y, \mathcal{F}(X))$ and $(X, Z, \mathcal{F}(X))$ has τ - $\mathcal{F}_{cmc}$ -SACP, $Y + Z$ is closed but $(X, Y + Z, \mathcal{F}(X))$ does not have τ - $\mathcal{F}_{cmc}$ -SACP.

Figure 2: Remark 3.12 of [1].

2 The compactness forced by the plateau gauge

Lemma 1. Let V be a subspace of X .

1. If $(X, V, \mathcal{F}(X))$ has norm- $\mathfrak{F}_{cmc}$ -SACP, then V is finite dimensional.
2. If $(X, V, \mathcal{F}(X))$ has weak- $\mathfrak{F}_{cmc}$ -SACP, then V is reflexive.

Proof. Let $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be

$$g(t) = \max\{t - 1, 0\}.$$

This g belongs to $\mathfrak{F}_{cmc}$: it is the maximum of two affine functions, hence convex; it is increasing; and $g(t) \rightarrow \infty$ as $t \rightarrow \infty$.

Apply the SACP hypothesis to the one-point finite set $F = \{0\}$. Since $0 \in V$,

$$\text{rad}_V^g(F) = \inf_{v \in V} g(\|v\|) = 0.$$

Moreover, for every $v \in B_V$, $g(\|v\|) = 0$. Hence every sequence in the closed unit ball B_V is an exact minimizing sequence for g and F .

If the topology is the norm topology, SACP says that every sequence in B_V has a norm-convergent subsequence. The norm topology is metric, so B_V is norm compact. By Riesz's lemma, the closed unit ball of a Banach space is norm compact only in finite dimension. Thus V is finite dimensional.

If the topology is the weak topology, SACP says that every sequence in B_V has a weakly convergent subsequence in X . The set B_V is weakly closed in X , since V is a closed linear subspace and the unit ball is weakly closed. Thus B_V is weakly sequentially compact. The weak topology on V agrees with the topology inherited from the weak topology of X , by Hahn–Banach extension of functionals. Eberlein–Smulian therefore implies that B_V is weakly compact in V . Hence V is reflexive. $\square$

3 Answer to Remark 3.12

Theorem 1. *Let τ be either the norm topology or the weak topology. Let Y and Z be subspaces of a Banach space X such that $Y + Z$ is closed. If $(X, Y, \mathcal{F}(X))$ and $(X, Z, \mathcal{F}(X))$ both have $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$, then $(X, Y + Z, \mathcal{F}(X))$ has $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$.*

Proof. First suppose that τ is the norm topology. By Lemma 1, Y is finite dimensional. Theorem 3.10 of [1] states that if Y is finite dimensional and $(X, Z, \mathcal{F}(X))$ has $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$, then $(X, Y + Z, \mathcal{F}(X))$ has $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$. This gives the conclusion.

Now suppose that τ is the weak topology. By Lemma 1, Y is reflexive. Theorem 3.14 of [1] states that if Y is a reflexive subspace, $Y + Z$ is closed, and $(X, Z, \mathcal{F}(X))$ has weak- $\mathfrak{F}_{cmc}\text{-SACP}$, then $(X, Y + Z, \mathcal{F}(X))$ has weak- $\mathfrak{F}_{cmc}\text{-SACP}$. This gives the conclusion in the weak case. $\square$

for all $\kappa \leq \max\{\nu_1, \nu_2\}$ and this concludes our assertion. $\square$

Theorem 3.10. *Let Y be a finite dimensional subspace of X , and Z be a subspace of X such that $(X, Z, \mathcal{F}(X))$ has $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$. Then, $(X, Y + Z, \mathcal{F}(X))$ has $\tau\text{-}\mathfrak{F}_{cmc}\text{-SACP}$.*

Proof. Let $F = \{x_1, \dots, x_N\} \in \mathcal{F}(X)$, $f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+$ be a function in $\mathfrak{F}_{cmc}$. Suppose $(y_n) \subseteq Y$, $(z_n) \subseteq Z$ be such that $r_f(y_n + z_n, F) \rightarrow \text{rad}_{Y+Z}^f(F)$.

Figure 3: The finite-dimensional transfer theorem, Theorem 3.10 of [1].

Theorem 3.14. *Let Y be a reflexive subspace of X and Z be a subspace of X such that $Y + Z$ is closed and $(X, Z, \mathcal{F}(X))$ has weak- $\mathfrak{F}_{cmc}\text{-SACP}$. Then, $(X, Y + Z, \mathcal{F}(X))$ has weak- $\mathfrak{F}_{cmc}\text{-SACP}$.*

Proof. Let $F = \{x_1, \dots, x_N\} \in \mathcal{F}(X)$, $f : \mathbb{R}_+^N \rightarrow \mathbb{R}_+$ be a function in $\mathfrak{F}_{cmc}$. Let (y_n) be a sequence in Y and (z_n) be a sequence in Z such that $r_f(y_n + z_n, F) \rightarrow \text{rad}_{Y+Z}^f(F)$. Without loss of generality, we may assume that $(r_f(y_n + z_n, F))$ is a decreasing sequence converging to $\text{rad}_{Y+Z}^f(F)$. Since f is coercive, $(y_n + z_n)$ is bounded, and from Theorem 3.4, both (y_n)

Figure 4: The weak/reflexive transfer theorem, Theorem 3.14 of [1].

Remark 1. *The proof uses the full function class $\mathfrak{F}_{cmc}$ exactly as defined in [1]. If one replaced $\mathfrak{F}_{cmc}$ by a narrower class, for example by requiring strict increase or requiring the gauge to vanish only at the origin, then the plateau-gauge argument would no longer apply. That would be a different question from Remark 3.12 as stated.*

4 Novelty check

The run's local indexes were searched for arXiv:2505.10851, the exact title, the exact phrase from Remark 3.12, and the keywords $\mathfrak{F}_{cmc}$, SACP, weak- $\mathfrak{F}_{cmc}\text{-SACP}$, and $Y + Z$. No duplicate solution

packet or literature-answer packet was found.

External web/arXiv searches on 2026-06-19 for the exact remark phrase, “Transfer of Approximation properties under Local Constraints Remark 3.12”, “F_cmc simultaneous approximative SACP subspaces”, and “weak-F_cmc-SACP reflexive subspace $Y+Z$ ” found the source paper [1] and the earlier SACP paper [2], but no later paper explicitly answering this subspace-sum question.

5 Human review recommendation

This should be reviewed as a short claimed full solution. The proof has only one new observation, namely that the plateau gauge $g(t) = \max\{t - 1, 0\}$ is an allowed element of $\mathfrak{F}_{cmc}$ and converts the whole unit ball of a subspace into exact minimizing sequences for $F = \{0\}$. Once that is accepted, the source paper’s Theorems 3.10 and 3.14 close the norm and weak cases.

References

- [1] Syamantak Das, *Transfer of Approximation properties under Local Constraints and Best Simultaneous Approximation on Sums*, arXiv:2505.10851v3, 2026. <https://arxiv.org/abs/2505.10851>
- [2] Syamantak Das and Tanmoy Paul, *A study on $\mathfrak{F}$ -simultaneous approximative τ -compactness property in Banach spaces*, arXiv:2501.03347, 2025. <https://arxiv.org/abs/2501.03347>
- [3] Robert E. Megginson, *An Introduction to Banach Space Theory*, Graduate Texts in Mathematics 183, Springer, 1998.